Case study #1 Duolingo

Improving Activation for Duolingo

Overview

Duolingo is facing a common problem:

> A large % of users sign up but **never complete their first meaningful learning milestone**.
>
> The team believes improving **activation (early success)** will drive long-term retention.

Data Snapshot (Given to Students)

Assume 10,000 users signed up.

Table 1

Step	Users	Conversion
App Installed	10,000	—
Account Created	8,500	85%
Language Selected	7,200	85%
Onboarding Completed	5,000	69%
First Lesson Started	4,200	84%
First Lesson Completed	2,800	67%
Day 2 Return	1,200	43%

Behavioral Observations

- Users who complete onboarding in **<2 minutes** are **2x more likely** to complete the first lesson
- Users who start a lesson within **5 minutes of signup** have **+60% higher retention**
- Mobile users drop off more during onboarding vs desktop
- Paid users have higher signup rates but **lower Day 2 retention**

1. Funnel Insights

Drop-off 1: Onboarding → First Lesson Completion (44%)

Framework: Friction vs Motivation (Behavioral Model)

Interpretation: Users have **intent (they onboard)** But lack **momentum to act immediately**

Root Cause:

- Cognitive overload (too many steps before action)
- No instant reward loop
- "I'll do it later" effect

Insight: Activation fails when **action is not immediate**

Drop-off 2: First Lesson → Day 2 Retention (57%)

Framework: Habit Loop (Cue → Action → Reward)

Interpretation: Users complete lesson but reward is not strong enough to create a loop

Root Cause:

- Weak emotional payoff
- No strong “reason to return”

Insight: Activation fails when value is not memorable

2. Activation Metrics

Activation is the moment when a user first experiences the core value of a product.

Chosen Metric: Day 2 Return (after completing first lesson)

Activation is not just about *doing something once*—it’s about experiencing value and choosing to come back. The real goal is daily learning habits, not one-time usage.

- First lesson completed = user tried the product
- Day 2 return = user found it valuable enough to build a habit

Why this matters: A user is truly activated when:

1. They experience value
2. They choose to come back

3. Key Insights

1. Momentum drives success, delay kills it

Users who start learning within 5 minutes have much higher retention.
=> **Successful users act immediately; unsuccessful users postpone and drop.**

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2. Lower friction = higher activation

Users completing onboarding faster (<2 min) are 2x more likely to finish the first lesson.
=> **Successful users face less friction; unsuccessful users get slowed down or overwhelmed.**

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3. Experience quality differs by context (mobile vs others)

Mobile users drop more during onboarding.
=> **Successful users likely have smoother, focused experiences; unsuccessful users face friction in distracted/mobile environments.**

Users who experience fast, frictionless,and immediate value activate—others drop before value is felt.

4. Recommendations

1. **Instant Lesson Start**

Problem: Users drop between onboarding and first lesson

Insight: Users who start quickly have higher retention

Solution: Auto-start first lesson immediately after onboarding (no extra step)

Expected Impact:
↑ First lesson completion
↑ Activation rate

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2. **Strong Post-Lesson Hook**

Problem: Low Day 2 return

Insight: Users don't feel strong value or reason to come back

Solution: After lesson: show progress + streak + next-day goal
("You learned 7 words! Come back tomorrow to continue ")

Expected Impact:
↑ Day 2 retention
↑ Habit formation

Experiment (A/B Test)

Hypothesis: Reducing delay between onboarding and learning increases activation

Test:
A: Current flow
B: Auto-start lesson flow

Metrics:

- First lesson completion
- Day 2 retention

Expected Result: Higher activation in variant B

5. Dashboard Thinking

North Star: **Day 2 Retention** (after first lesson)

Why:

- Best indicator of true activation (value + return intent)
- Directly linked to long-term retention and growth

Supporting Metrics:

1. **First Lesson Completion Rate**: Measures if users are actually experiencing initial value

2. **Time to First Lesson Start**: Measures onboarding friction (how quickly users reach value)

on

Insight: Users who start quickly have higher retention

Guardrail Metric:

Day 7 Retention

- Ensures we’re not optimizing only short-term activation
- Confirms users are building a sustainable habit, not just returning once

6. Startup analytics timing

- Start investing in analytics once MVP is live and users are interacting with the product
- As soon as you see users dropping or behaving unexpectedly, you need data to understand why
- Begin with basic funnel + retention tracking (don't overcomplicate early)
- Invest deeper when:
    - You're making product decisions based on user behavior
    - You're starting to scale acquisition
- Avoid vanity metrics (downloads); focus on activation and retention